sensors

MDPI

Supplementary Materials

Phase-Robust Active Self-Test for Fault Localization in Mixed-Signal Sensor Interfaces: A Simulation Study — Supplementary Materials

Moahaimen Talib

This document provides the detailed results that support the main article. All values are simulation results recomputed from the saved per-record predictions of the three validation dataset seeds (20261026, 20261027 and 20261028) with the aggregation and bootstrap procedure described in Section 6.5 of the main article: confusion matrices of the pooled cells are summed per seed and initialization, the metric is averaged over two initializations and three seeds, and intervals are paired condition-level cluster-bootstrap 95% intervals with 5000 resamples per seed. Abbreviations are as in the main article.

S1. Fine-Fault Confusion of the Primary Method

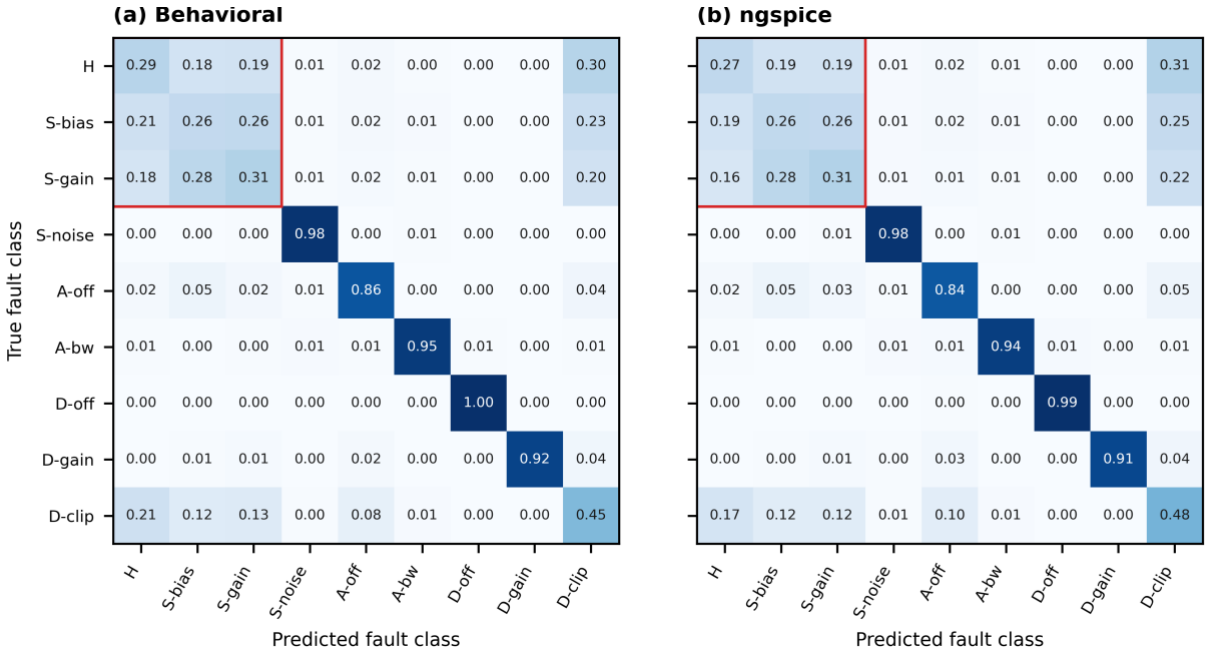


Figure S1. Row-normalized fine-fault confusion matrices of PRr with the monitor at $E = \pm 2\%$, pooled over amplitudes and the five phases and averaged over three dataset seeds × two initializations, in (**a**) the behavioral and (**b**) the ngspice domain (simulation results). H, healthy; S, sensor; A, AFE; D, ADC. The red box marks healthy, sensor bias and sensor gain; together with ADC clipping these classes form the group that the references do not separate (Section 7.7 of the main article).

https://doi.org/10.3390/xxxxx

S2. Supplementary Tables

Table S1. Layer macro-F1 per edge phase for every arm, pooled over each domain's amplitudes (behavioral ×0.8–×1.2, ngspice ×0.8, ×1.0, ×1.2). Spread = largest minus smallest of the five phases.

Domain	Arm	Reference condition	ψ = 0	ψ = 0.25	ψ = 0.5	ψ = 0.75	ψ = 0.999	Spread
Behavioral	A0	Monitor, E = ±2%	0.279	0.279	0.281	0.281	0.282	0.003
Behavioral	A0	Known levels	0.279	0.279	0.281	0.281	0.282	0.003
Behavioral	FS	Monitor, E = ±2%	0.649	0.636	0.614	0.590	0.568	0.081
Behavioral	FS	Known levels	0.638	0.625	0.603	0.580	0.556	0.082
Behavioral	FSr	Monitor, E = ±2%	0.651	0.650	0.648	0.641	0.626	0.025
Behavioral	PR	Monitor, E = ±2%	0.657	0.656	0.655	0.655	0.653	0.004
Behavioral	PRr	Monitor, E = ±2%	0.659	0.658	0.656	0.653	0.652	0.006
Behavioral	PRr	Known levels	0.648	0.646	0.646	0.645	0.643	0.005
Behavioral	PRrM	Monitor, E = ±2%	0.677	0.676	0.672	0.668	0.667	0.010
Behavioral	PRrM	Known levels	0.690	0.687	0.686	0.683	0.679	0.011
ngspice	FS	Monitor, E = ±2%	0.642	0.625	0.606	0.584	0.561	0.081
ngspice	FS	Known levels	0.634	0.618	0.597	0.572	0.550	0.085
ngspice	FSr	Monitor, E = ±2%	0.643	0.642	0.638	0.631	0.617	0.026
ngspice	PR	Monitor, E = ±2%	0.649	0.648	0.647	0.644	0.642	0.007
ngspice	PRr	Monitor, E = ±2%	0.654	0.651	0.649	0.645	0.644	0.010
ngspice	PRr	Known levels	0.645	0.640	0.641	0.637	0.634	0.011
ngspice	PRrM	Monitor, E = ±2%	0.667	0.665	0.664	0.659	0.658	0.009
ngspice	PRrM	Known levels	0.683	0.682	0.677	0.674	0.670	0.013

Table S2. Layer macro-F1, retention relative to N_d, AFE and ADC recall and AFE–ADC cross-confusion of PRr at ±20% reference amplitude for every reference condition, pooled over the five phases.

Domain	Reference condition	Amplitude	Layer macro-F1	Retention	AFE recall	ADC recall	Cross-confusion κ
Behavioral	Known levels	×0.8	0.615	0.934	0.912	0.831	0.026
Behavioral	Known levels	×1.2	0.643	0.976	0.898	0.797	0.089
Behavioral	Monitor, E = 0%	×0.8	0.610	0.926	0.883	0.843	0.035
Behavioral	Monitor, E = 0%	×1.2	0.642	0.975	0.865	0.797	0.085
Behavioral	Monitor, E = ±1%	×0.8	0.608	0.923	0.905	0.833	0.032
Behavioral	Monitor, E = ±1%	×1.2	0.644	0.978	0.876	0.790	0.087
Behavioral	Monitor, E = ±2%	×0.8	0.614	0.933	0.886	0.826	0.035
Behavioral	Monitor, E = ±2%	×1.2	0.663	1.006	0.885	0.797	0.079
Behavioral	Monitor, E = ±5%	×0.8	0.611	0.928	0.891	0.778	0.033
Behavioral	Monitor, E = ±5%	×1.2	0.649	0.985	0.878	0.777	0.083
ngspice	Known levels	×0.8	0.616	0.933	0.909	0.833	0.028
ngspice	Known levels	×1.2	0.642	0.973	0.898	0.799	0.088
ngspice	Monitor, E = 0%	×0.8	0.606	0.918	0.879	0.845	0.036
ngspice	Monitor, E = 0%	×1.2	0.644	0.975	0.863	0.800	0.083
ngspice	Monitor, E = ±1%	×0.8	0.604	0.916	0.899	0.835	0.033
ngspice	Monitor, E = ±1%	×1.2	0.644	0.976	0.874	0.792	0.086
ngspice	Monitor, E = ±2%	×0.8	0.611	0.925	0.883	0.832	0.036

Domain	Reference condition	Amplitude	Layer macro-F1	Retention	AFE recall	ADC recall	Cross-confusion κ
ngspice	Monitor, E = ±2%	×1.2	0.664	1.006	0.885	0.797	0.078
ngspice	Monitor, E = ±5%	×0.8	0.610	0.924	0.887	0.773	0.034
ngspice	Monitor, E = ±5%	×1.2	0.646	0.979	0.875	0.778	0.081
Behavioral	No monitor (nominal levels assumed)	×0.8	0.310	0.471	0.426	0.944	0.230
Behavioral	No monitor (nominal levels assumed)	×1.2	0.237	0.360	0.375	0.820	0.336

Table S3. Recall per fault class and per layer of every arm with the monitor at E = ±2%, pooled over amplitudes and phases. Bh., behavioral; ng., ngspice; H, healthy; S, sensor; A, AFE; D, ADC; bw, bandwidth; clip, clipping. The last row is the known-level PRr network applied without monitoring at ×0.8 and ×1.2.

Dom.	Arm	H	S-bias	S-gain	S-noise	A-offset	A-bw	D-offset	D-gain	D-clip	H layer	S layer	AFE layer	ADC layer
Bh.	A0	0.01	0.05	0.06	0.09	0.13	0.51	0.54	0.80	0.06	0.01	0.15	0.24	0.88
Bh.	FS	0.23	0.23	0.31	0.99	0.76	0.94	0.99	0.95	0.40	0.10	0.70	0.87	0.82
Bh.	FSr	0.33	0.32	0.26	0.98	0.82	0.93	1.00	0.96	0.31	0.15	0.80	0.87	0.76
Bh.	PR	0.28	0.28	0.32	0.98	0.86	0.96	1.00	0.92	0.43	0.13	0.78	0.93	0.80
Bh.	PRr	0.29	0.26	0.31	0.98	0.86	0.95	1.00	0.92	0.45	0.14	0.78	0.91	0.81
Bh.	PRrM	0.36	0.26	0.34	0.98	0.92	0.94	1.00	0.93	0.37	0.24	0.74	0.94	0.78
ng.	FS	0.23	0.24	0.31	0.99	0.75	0.92	0.99	0.94	0.38	0.09	0.69	0.86	0.81
ng.	FSr	0.31	0.33	0.26	0.99	0.81	0.91	1.00	0.95	0.30	0.14	0.80	0.87	0.76
ng.	PR	0.25	0.28	0.33	0.98	0.84	0.94	0.99	0.90	0.45	0.12	0.77	0.92	0.80
ng.	PRr	0.27	0.26	0.31	0.98	0.84	0.94	0.99	0.91	0.48	0.13	0.77	0.90	0.81
ng.	PRrM	0.32	0.26	0.35	0.98	0.91	0.93	1.00	0.91	0.40	0.22	0.74	0.93	0.78
Bh.	PRr, no monitor	–	–	–	–	–	–	–	–	–	0.00	0.04	0.40	0.88

Table S4. All 70 comparisons of gate B (behavioral domain, pooled over the five phases): PRr minus the baseline in layer macro-F1 with cluster-bootstrap 95% intervals. The baseline networks with PRr descriptors are trained for the same reference condition as PRr.

Reference condition	Amplitude	Two-point, A0	Passive, A0	Single probe, A0	AFE-only, A0	ADC-only, A0	AFE-only, PRr descriptors	ADC-only, PRr descriptors
Known levels	×0.8	+0.475 [+0.461, +0.489]	+0.356 [+0.341, +0.371]	+0.266 [+0.250, +0.282]	+0.295 [+0.280, +0.311]	+0.495 [+0.482, +0.508]	+0.142 [+0.128, +0.155]	+0.160 [+0.144, +0.175]
Known levels	×1.2	+0.521 [+0.507, +0.535]	+0.384 [+0.368, +0.400]	+0.246 [+0.230, +0.261]	+0.348 [+0.329, +0.366]	+0.521 [+0.507, +0.534]	+0.126 [+0.112, +0.140]	+0.190 [+0.176, +0.206]
Monitor, E = 0%	×0.8	+0.470 [+0.456, +0.483]	+0.351 [+0.337, +0.366]	+0.261 [+0.246, +0.276]	+0.290 [+0.276, +0.305]	+0.490 [+0.478, +0.502]	+0.134 [+0.122, +0.145]	+0.154 [+0.140, +0.168]
Monitor, E = 0%	×1.2	+0.521 [+0.508, +0.533]	+0.383 [+0.369, +0.399]	+0.246 [+0.231, +0.260]	+0.347 [+0.329, +0.365]	+0.521 [+0.508, +0.533]	+0.127 [+0.114, +0.141]	+0.185 [+0.171, +0.199]

Reference condition	Amplitude	Two-point, A0	Passive, A0	Single probe, A0	AFE-only, A0	ADC-only, A0	AFE-only, PRr descriptors	ADC-only, PRr descriptors
Monitor, E = ±1%	×0.8	+0.468 [+0.455, +0.480]	+0.349 [+0.335, +0.364]	+0.259 [+0.244, +0.274]	+0.288 [+0.274, +0.303]	+0.488 [+0.477, +0.499]	+0.135 [+0.123, +0.147]	+0.154 [+0.140, +0.168]
Monitor, E = ±1%	×1.2	+0.523 [+0.510, +0.535]	+0.385 [+0.371, +0.401]	+0.247 [+0.233, +0.262]	+0.349 [+0.331, +0.368]	+0.522 [+0.510, +0.534]	+0.133 [+0.120, +0.147]	+0.190 [+0.175, +0.205]
Monitor, E = ±2%	×0.8	+0.474 [+0.459, +0.488]	+0.356 [+0.340, +0.371]	+0.265 [+0.250, +0.281]	+0.294 [+0.279, +0.310]	+0.494 [+0.481, +0.507]	+0.144 [+0.131, +0.157]	+0.153 [+0.138, +0.167]
Monitor, E = ±2%	×1.2	+0.541 [+0.527, +0.555]	+0.404 [+0.389, +0.420]	+0.266 [+0.251, +0.282]	+0.368 [+0.348, +0.388]	+0.541 [+0.528, +0.554]	+0.156 [+0.141, +0.170]	+0.203 [+0.187, +0.219]
Monitor, E = ±5%	×0.8	+0.471 [+0.455, +0.488]	+0.353 [+0.336, +0.370]	+0.263 [+0.246, +0.279]	+0.292 [+0.275, +0.309]	+0.492 [+0.477, +0.506]	+0.144 [+0.130, +0.159]	+0.164 [+0.148, +0.180]
Monitor, E = ±5%	×1.2	+0.528 [+0.512, +0.543]	+0.390 [+0.374, +0.407]	+0.252 [+0.236, +0.269]	+0.354 [+0.334, +0.373]	+0.527 [+0.513, +0.541]	+0.138 [+0.122, +0.154]	+0.199 [+0.182, +0.216]

Table S5. Computational cost and run time. Timings are desktop measurements of a Python implementation.

Quantity	Value
MLP architecture	20 inputs, hidden layers of 96 and 48 ReLU units, 4 (layer) or 9 (fault) outputs
MLP parameters	6,868 (layer network); 7,113 (fault network)
Multiply–accumulate operations per inference	6,720 (layer); 6,960 (fault)
Memory for the layer network (32-bit floating point)	about 27 kB (derived from the parameter count)
MLP inference time	0.68 μs per record (batched, Python, mean of three seeds)
Descriptor extraction time	Phase-robust 0.79 ms; Revision-1 0.47 ms per record (Python, one core)
Training time of PRr networks	mean 0.62 s, maximum 1.49 s per network (180 networks)
ngspice runs	13,878 per seed; 41,634 in total
Run time, seed 20261026	total 1175.6 s (ngspice 908.1 s with 3 workers; behavioral 707.7 s)
Run time, seed 20261027	total 1179.0 s (ngspice 908.9 s with 3 workers; behavioral 708.5 s)
Run time, seed 20261028	total 1179.1 s (ngspice 905.9 s with 3 workers; behavioral 703.8 s)
Not measured	Microcontroller timing, memory footprint, energy, power and silicon area

Table S6. Software environment and integrity of the evidence files.

Item	Value
Python	3.12.12
NumPy	2.2.6
pandas	2.3.3
SciPy	1.16.3
scikit-learn	1.7.2
matplotlib (original run)	3.10.7
ngspice	ngspice-47

Item	Value
Platform	macOS 27.0.1, arm64
Source of the versions	Reproducibility checklist generated on 14 October 2026, the day of the validation run, on the same machine; installation dates of ngspice-47 (3 October 2026) and scikit-learn 1.7.2 precede the run
SHA-256 of PROTOCOL_PHASE_ROBUST_FROZEN .md	a2f0bdd44acb8952f1ae0fec8af1dee31d907f9fadf8cc8a12d1a62314051cee
SHA-256 of decision_rules.json	170dac9a1a12884f021d5ddc9524cd190795b2dad283ede8778e32b24ffa4047
SHA-256 of predictions.npz (seed_20261026)	c2ba32a5eb70640759a84b1e7324406acbe751a4eefa6c4b5a35bf0acf18da0a
SHA-256 of predictions.npz (seed_20261027)	22ab4b3965a98edc3edbb67f9f6a4a1d0d413ba214b25f084d19ce34be71fc71
SHA-256 of predictions.npz (seed_20261028)	e8c14eb9a9106bde124f979f2753a667417e9f59b2bf5399eaf3fb37b7338903
SHA-256 of source_hashes_at_freeze.txt	56d6b7a1705a55546841dfae2abaa541089358c0248e144d1defd07a6c1fe394

The SHA-256 hashes of all simulator and analysis source files at the protocol freeze are listed in source_hashes_at_freeze.txt; the current source files match them.